Discussion of Indexing on 704:

In general:

$$Ae = A \pm C(I)$$

To go thru an array: (a) Set initial value of index i_0

(b) advance index by Δ

(c) Test index to exit i_x

ambiguity as to what initial value is called (0 or 1)

forward

A	B
0	1
1	2
2	3
3	4
4	5
⋮	⋮

backward

A	B
⋮	⋮
4	5
3	4
2	3
1	2
0	1

Starting with 0 is more natural, but 1 is easier on machine.

$i_x =$ [illegible] $m+1$

Complements:

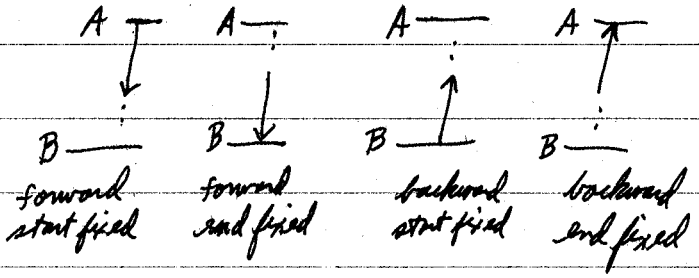
example $12 - 3 = 9_{10}$ $\quad$ $14 - 3 = 11_8$ $\quad$ $-12 + 3 = -9$

0011 +3
0010 +2
0001 +1
0000
1111 -1
1110 -2
1101 -3

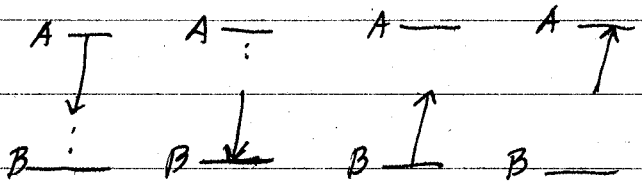
Neg = comp + 1

Eight cases are possible!

op address A



op address B



[illegible] -14 = 10011
+1
10100

+12 01100
-3 11101
+9 01001

-12 10100
+3 00011
10111
-9
-1
10110

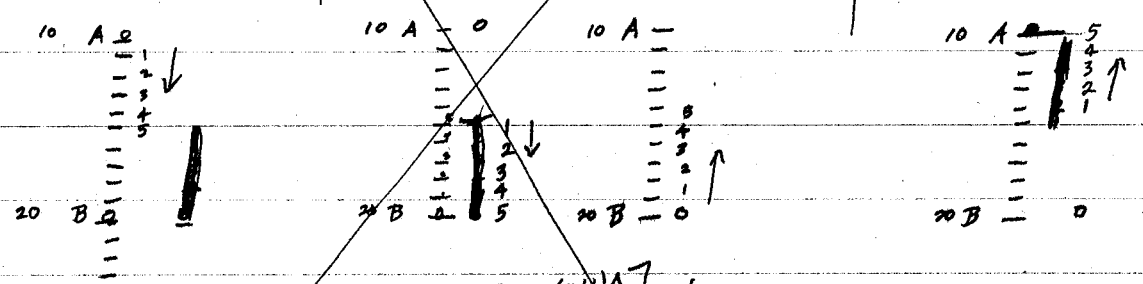
comp 01001

$B - m + 1 = A$

n = no. of values.

assuming array starts with 1 not zero: (call $B-A=m$)

for op Address A	forward start fixed $-(+\Delta)$	forward end fixed $-(+\Delta)$	backward start fixed $-(-\Delta)$	backward end fixed $-(-\Delta)$
i_0	Δ	$m-(n+1)\Delta$	$m-\Delta$	$n-1$
i_{L}	$n\Delta$	m	$m-n-1$	-1



LXD $\mathcal{L}(A), i$
α TXI $\alpha+1, i, -\Delta$
$\mathcal{L}+1$ TXL Back, i, $-(m\Delta)$

LXD $\mathcal{L}$ [illegible] $+m-(m+1)\Delta$, i
α TXI $\alpha+1, i, -\Delta$
$\mathcal{L}+1$ TXL Back, i, $-(m)$

LXD $\mathcal{L}$

TXI Y $C(i)+D \rightarrow C(i)$ Increment only uncond br to Y

TXH Y Test $\begin{cases} C(i) > D & \text{Tr to Y} \\ C(i) \leq D & \text{next instr} \end{cases}$ Test only

TXL Y Test $\begin{cases} C(i) \leq D & \text{Tr to Y} \\ C(i) > D & \text{next instr} \end{cases}$

TIX Y Test $\begin{cases} C(i) > D & C(i)-D \rightarrow C(i) \quad \text{Tr to Y} \\ C(i) \leq D & \text{next instr} \end{cases}$

TNX Y test $\begin{cases} C(i) \leq D & \text{Tr to Y} \\ C(i) > D & C(i)-D \rightarrow C(i) \quad \text{next instr} \end{cases}$

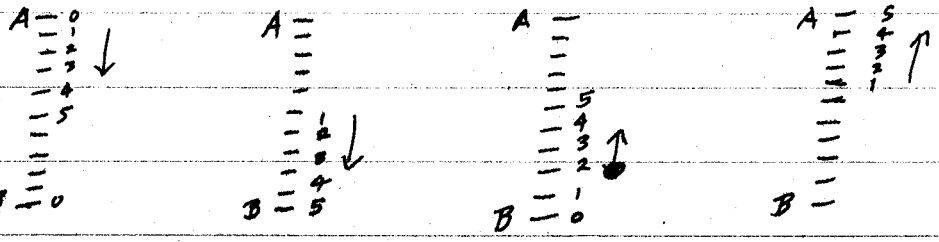
assuming array with non-zero start

n = no. of values

Δ = distance apart

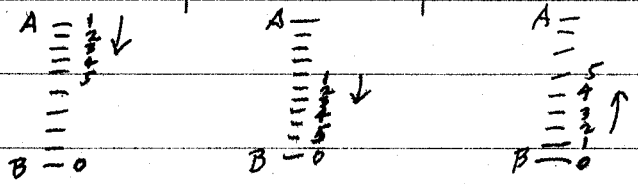
m = $B-A$ block size.

for op Addr A	stored forward start fixed $-(+\Delta)$	stored forward End fixed $-(+\Delta)$	stored backward Start fixed $-(-\Delta)$	stored backward End fixed $-(-\Delta)$
1st Value i_0	Δ	$m-(n+1)\Delta$	$m-\Delta$	$(n-1)\Delta$
last Value i_L	$n\Delta$	m	$m-n\Delta$	0



LXD $\mathcal{L}(\Delta)$, i	$\mathcal{L}[m-(n+1)\Delta]$	$m-\Delta$	$(n-1)\Delta$
α TXI $\alpha+1$, i, $-\Delta$	$-\Delta$	$+\Delta$	$+\Delta$
$\alpha+1$ TXL BACK, i, $-(n\Delta)$	$-m$	$-(m-n\Delta)$	0

for op Addr B	stored forward Start fixed $-(+\Delta)$	stored forward End fixed $+(+\Delta)$	stored backward Start fixed $-(-\Delta)$	stored backward End fixed
1st Value i_0	$-m$	$-n\Delta$	$-\Delta$	$-m+(n-1)\Delta$
last Value i_L	$-m+(n-1)\Delta$	$-\Delta$	$-n\Delta$	$-m$



LXD	$\mathcal{L}(-m)$, i	LXD $\mathcal{L}(-n\Delta)$, i	$\mathcal{L}(-\Delta)$, i
α TXI	$\alpha+1$, i, $-\Delta$	$-\Delta$	
$\alpha+1$ TXL	BACK, i, $m-(n-1)\Delta$	$-\Delta$	